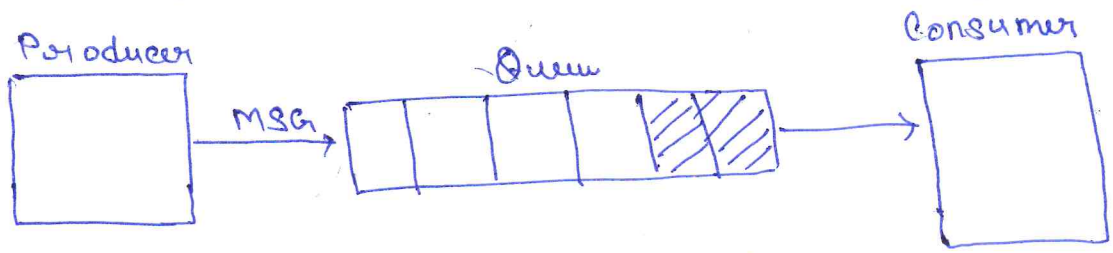
Distributed Messaging Queue

Lecture 53

1. What is Messaging Queue? Why its needed & its advantages?
2. What is Point 2 Point & Pub/Sub Messaging Types?
3. How Messaging Queue Works?
   - → KAFKA
   - → RABBIT MQ
4. What happen when Queue size limit is reached?
   we have add different broker.
5. What happens to Messages, when Queue goes down?
   we make followers to Leader.
6. What happens when Consumer goes down?
   another consumer take over
7. What happen when Consumer not able to process Msg?
8. How Retry works? Different ways to Retry?
9. How Distributed Messaging Queue works?
10. What is Dead Letter Queue?

Concept && Coding

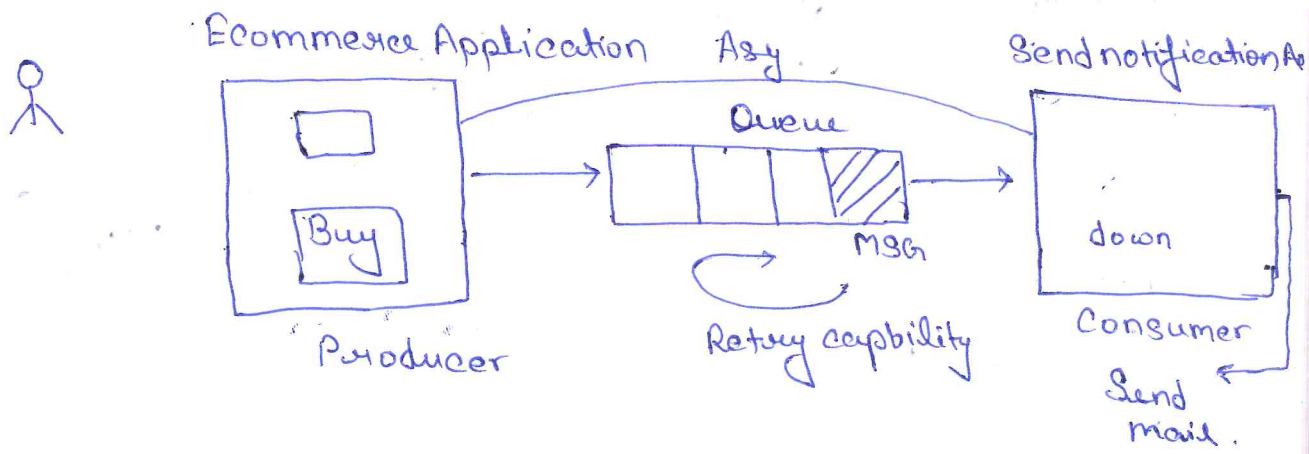
Messaging Queue & Why its Needed :→



Messaging Queue :- One have producer It have produces a message. This message goes on particular Queue. Consumer can read the messages.

Why its needed ?

1 UseCase :- Asynchronous Nature



* My producer is ecommerce application. User is Buying a particular product. After as soon as you buy it we need to send a notification. But user don't have to actually wait for that notification.

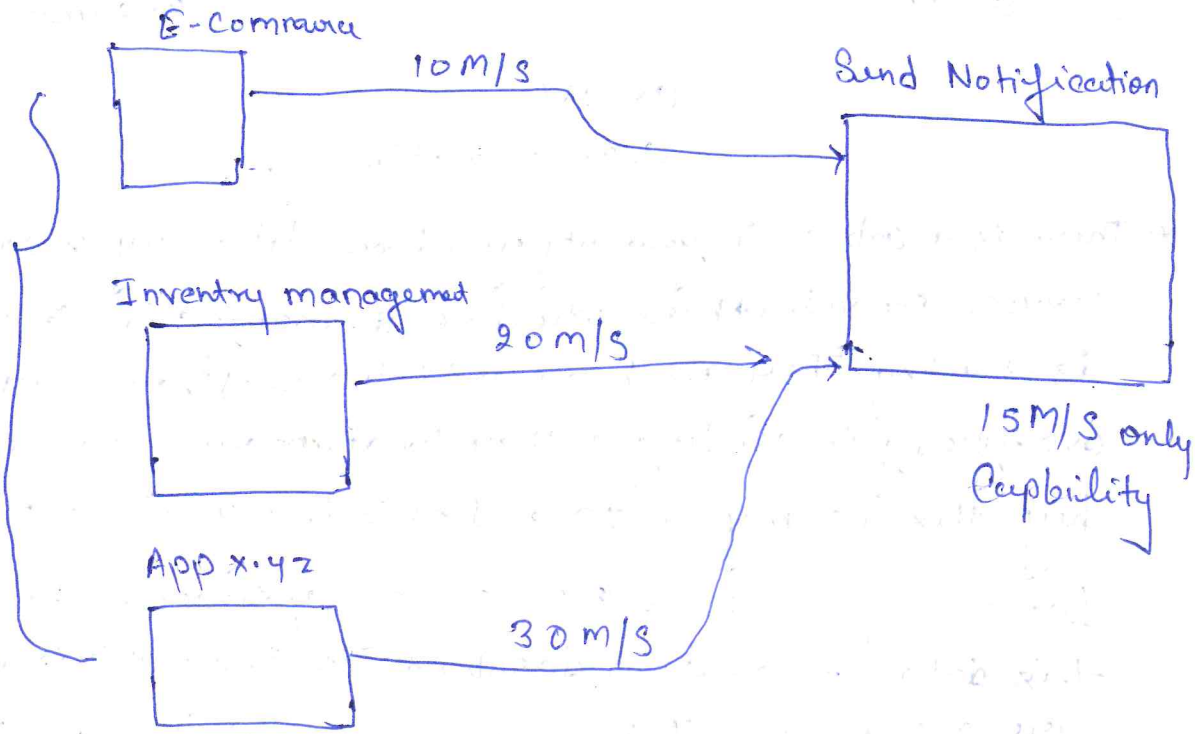
Concept && Coding

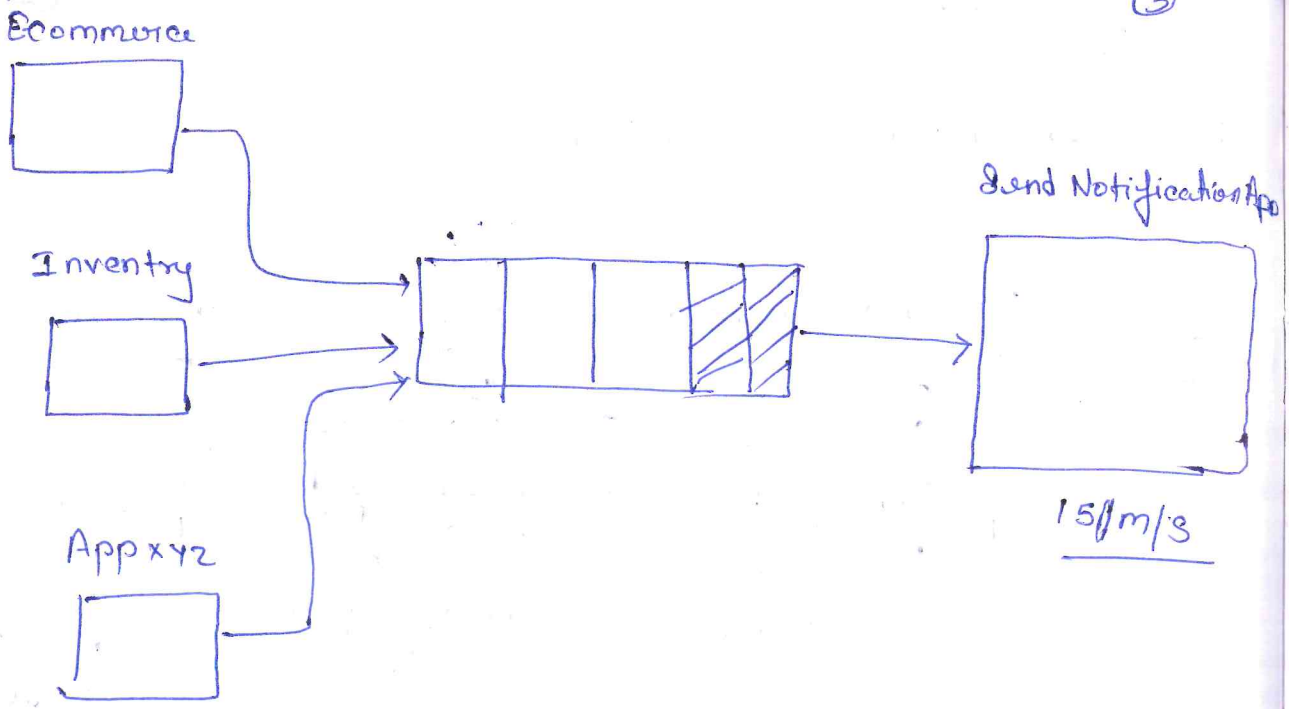
Advantages

1 :- Latency will be increase

2 :- Retry capbilities

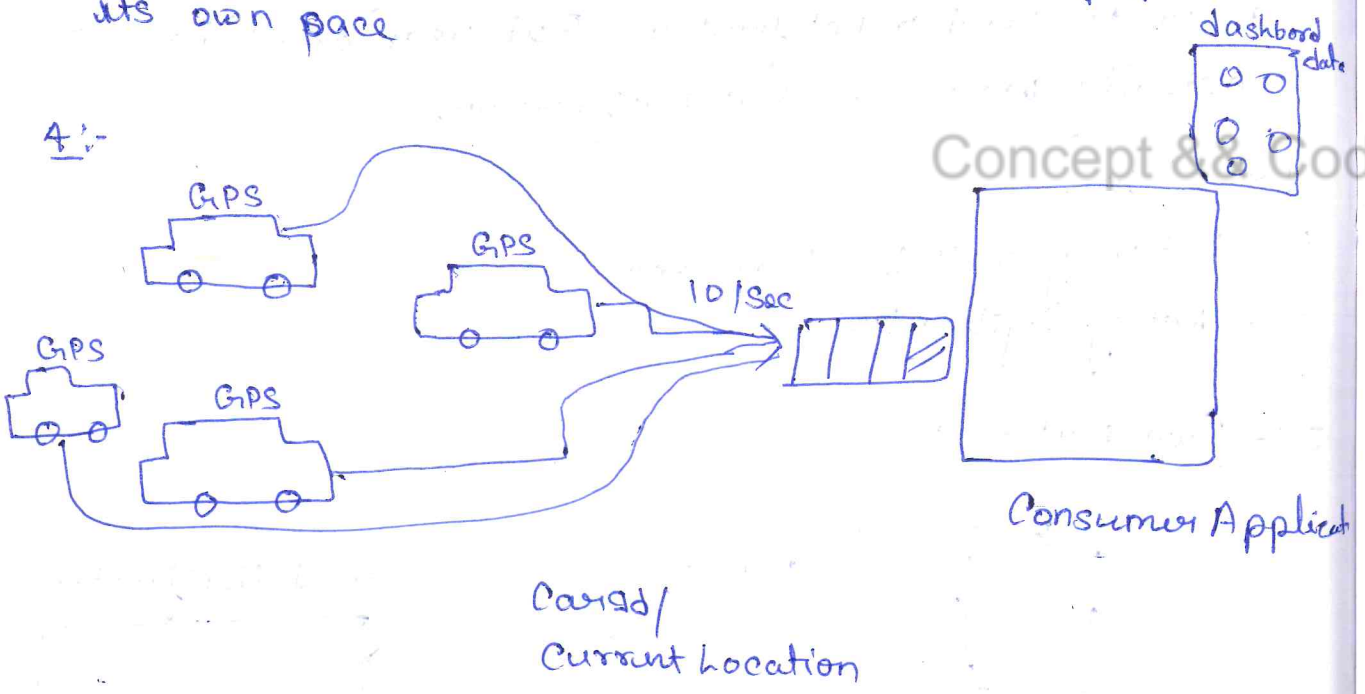
3 :- Pace Matching :-





* All three put all the message into a message Queue. and Subscribe the consumer will consume it from in its own pace

4:-

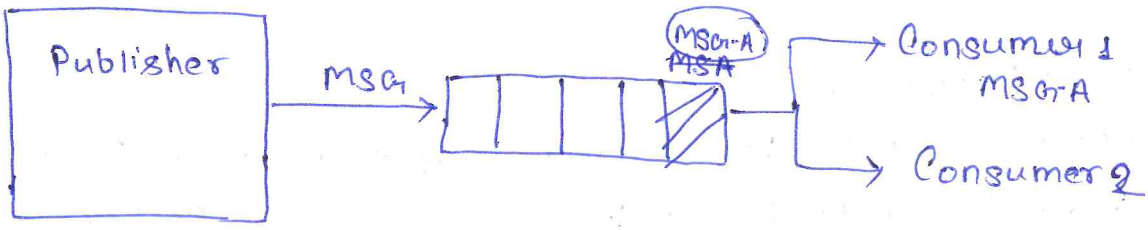


Concept && Coding

* There is a cab so in your city you have let's say a cab service provider those who have so many cabs. ~~Everybody is sending~~ They all have GPS installed, now let's say everybody is sending a their location every 10 second. hey this is my car ID and this is my current location. Let's say we have a consumer application. wich uses this data to create a dashbord like okay this is these are car location. This consumer application can not Que

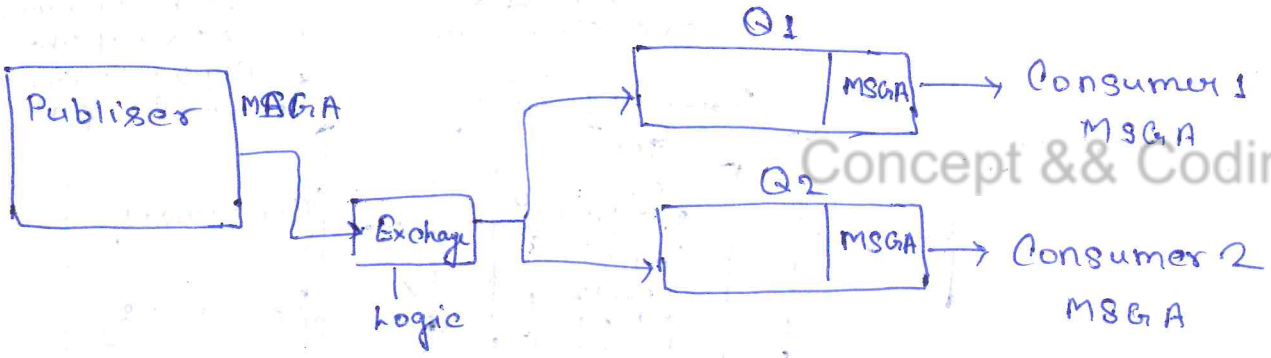
Point 2 point (P2P) && Pub|Sub

P2P :-



- Point 2 Point (P2P) messaging service only it will be consumed or processed only once. It can be be processed by either consumer but only once.

Pub/Sub :-



Concept && Coding

- Wheever a message comes in right based upon the logic It put the message it broadcasts the message to all the queues. And Both Consumer have consume and processed the message.

- How messaging Queue Works ?

1 Kafka :→

What are the Component involve

(1) • Producer

(2) • Consumer :- Consumer read the partition. Data will be stor

(3) • Consumer Group :- Each Consumer is part of Consumer Grou

(4) • Topic : Topic like a category or channel where dat store and organized.

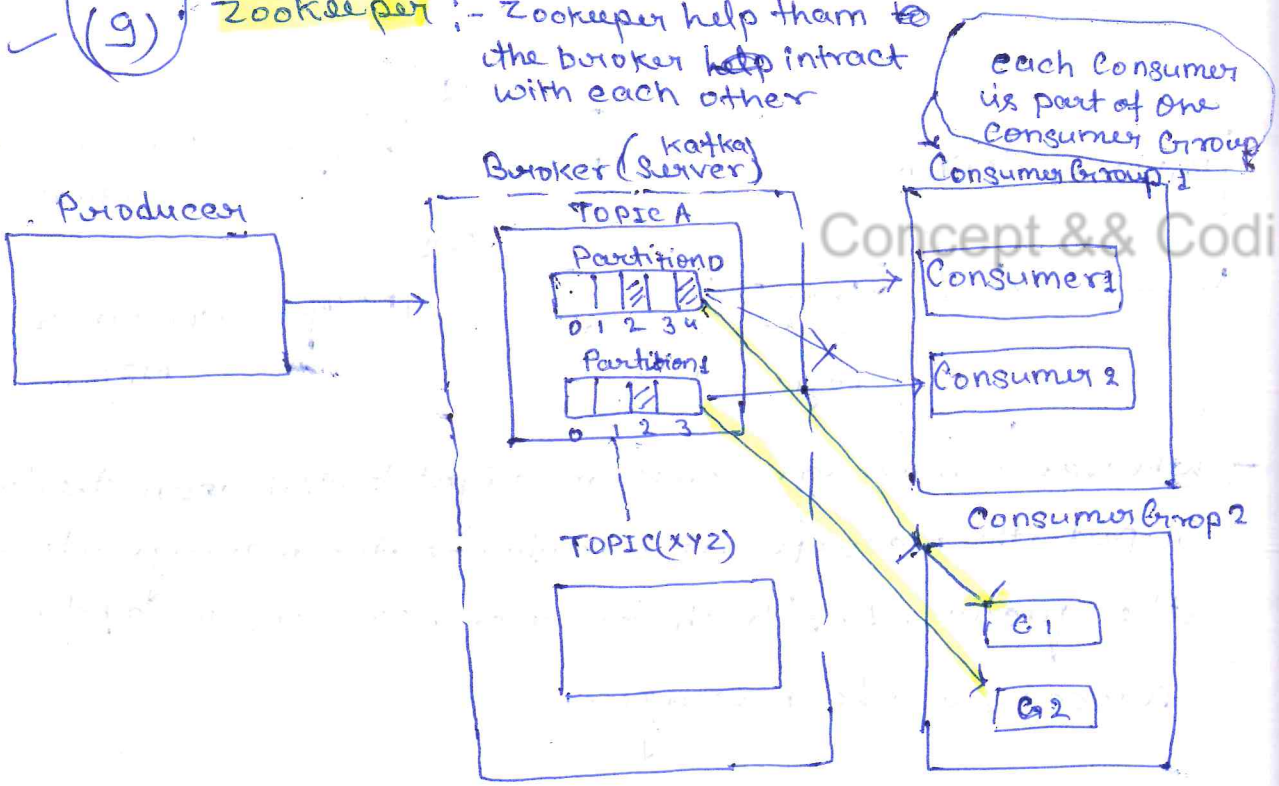
(5) • ~~Partion~~ Partition : Partition like small part of topic helps in distri

(6) • Offset :- Offset is nothing like a queue. different partition can

(7) • Broker :- Producer talk with Broker. It is kafka Server.

(8) • Cluster :- Group of Brokers called as cluster.

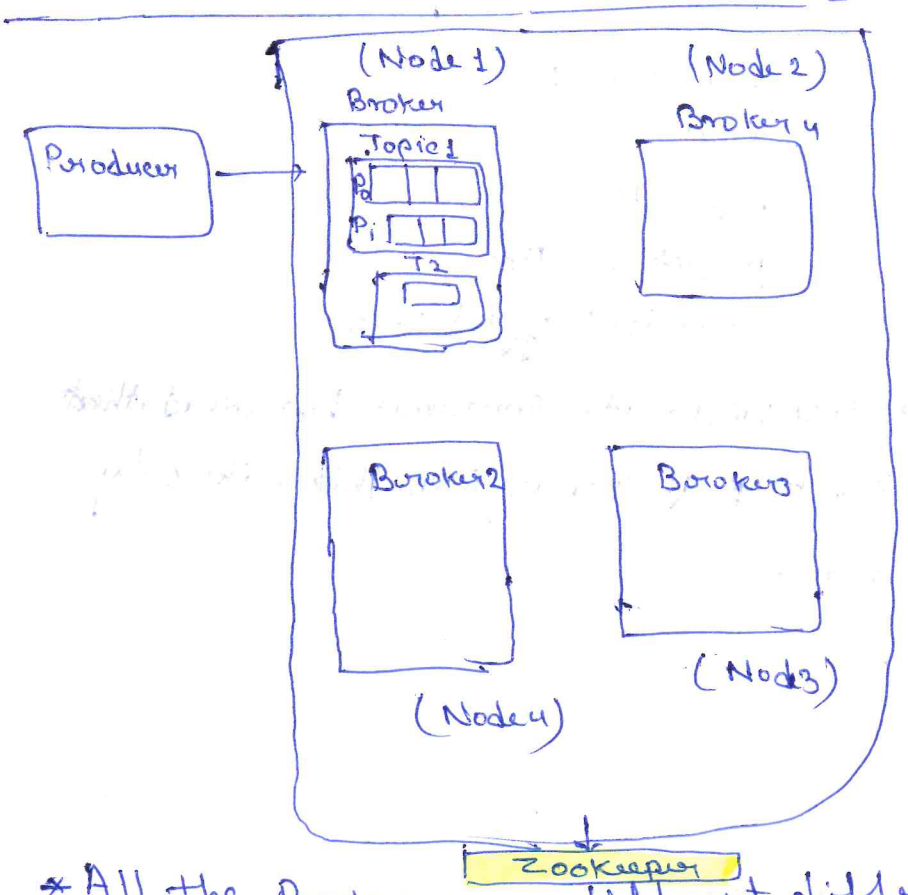
(9) • Zookeeper :- Zookeeper help tham to the broker ~~help~~ intract with each other



Concept && Coding

Note :- The Consumer inside a particular group might they have certain. Topic A have two partition, 'Partition 0' and partition 1. Inside this consumer Group 1. If Consumer1 have using partition 0. Consumer 2 will be take Partition 1. ~~But~~ But Consumer 2 will not take partition 0 It is not possible. Partition 0 have use different Consumer Group2

Store on partition only
Group.
grouped.
distributing and processing data efficiently.
can have different length.



Concept && Coding

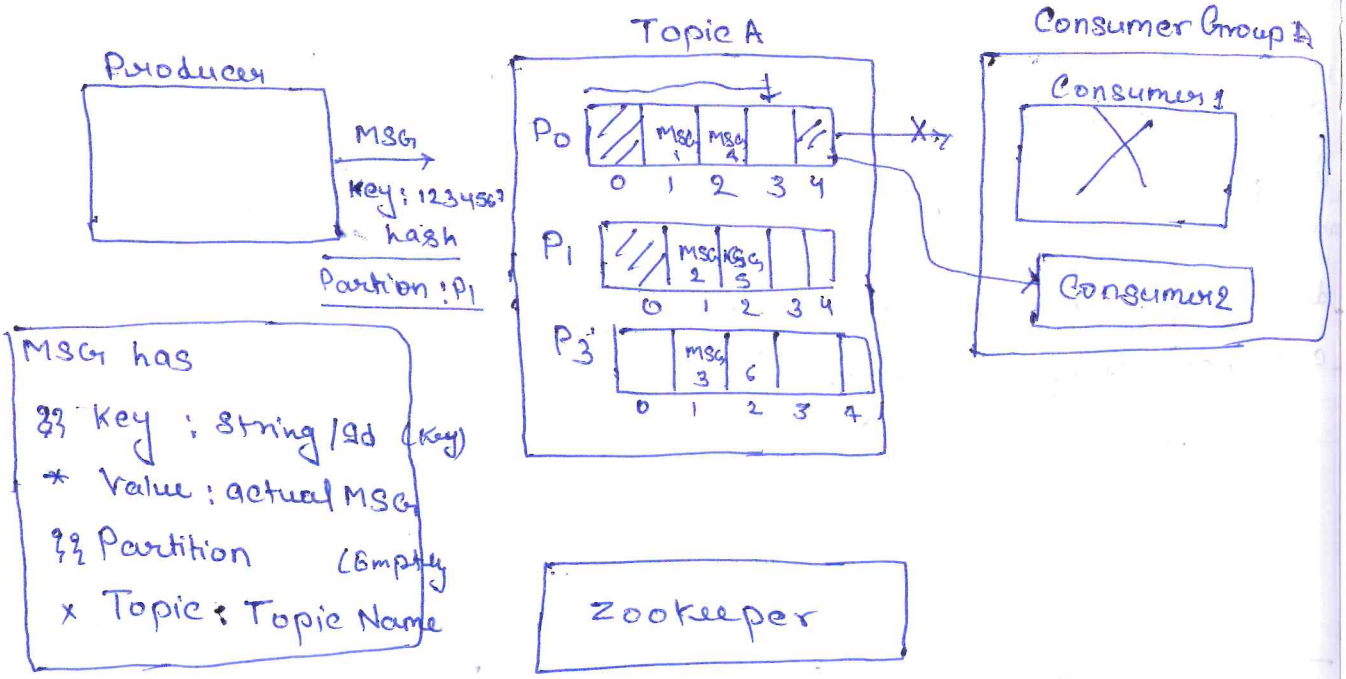
* All the Broker run different different machine

Group of Broker is called a Cluster.

* They will Interect with each other then we have something called Zookeeper.

- Zookeeper helps an internal communication that okay Topic 1 this partisan which broker.

How this message work ??



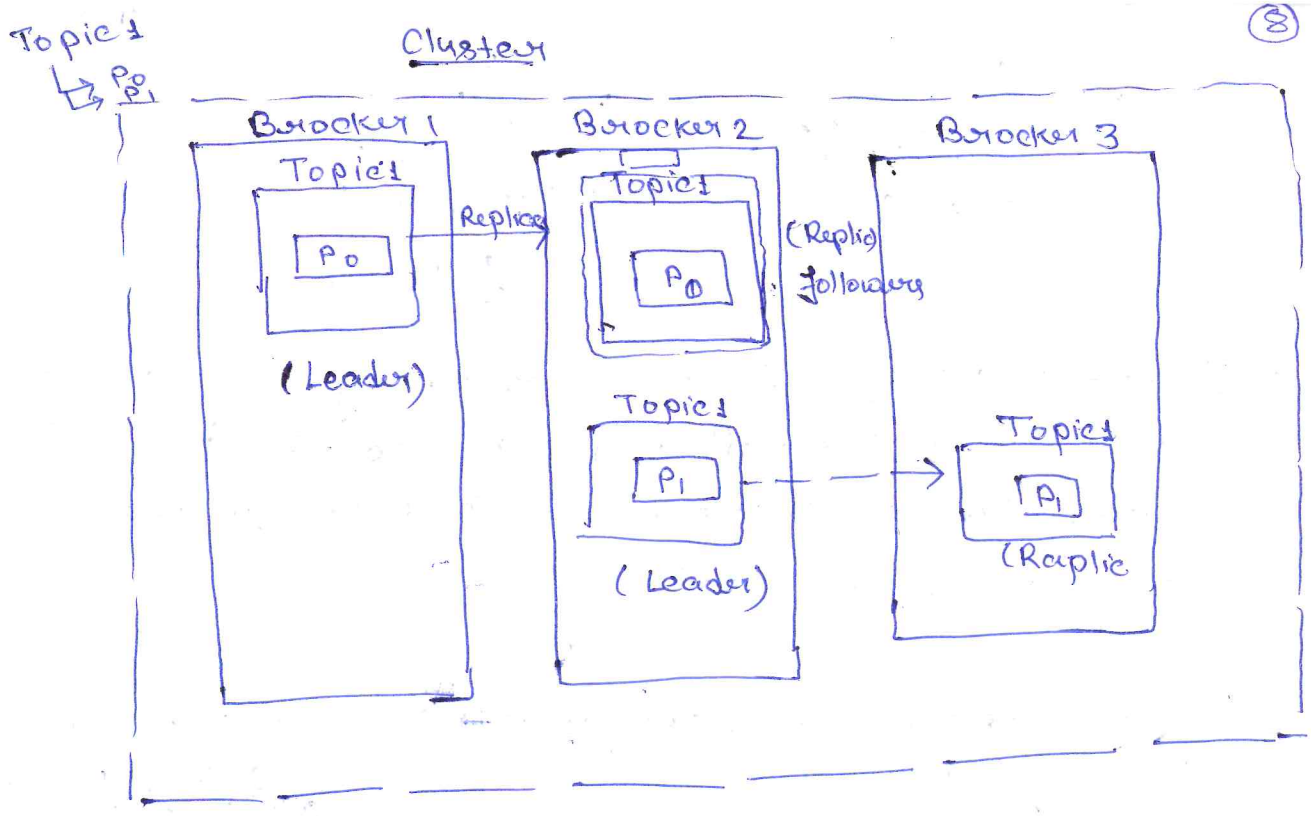
Zookeeper

Consumer 1
CG A
Topic A
Partition : P0
Commited offset :- 3

* use case :-

* offset tells that till where the consumer has read the message right so ~~keep~~ zookeeper keep this like only this the consumer.

Concept && Coding



* Brocker 1 has Topic 1 and partition 0 and Brocker 2 has Topic 1 and partition 1.

* even the partitions can reside into different Kafka servers.

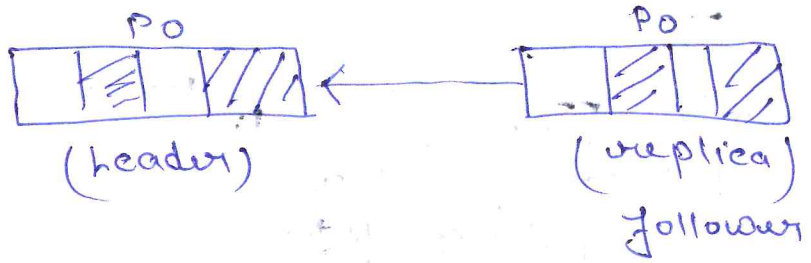
Concept && Coding

- What if partition goes down what will happen to the message which is inside?

Ans:- That's where we have a replica inside here.

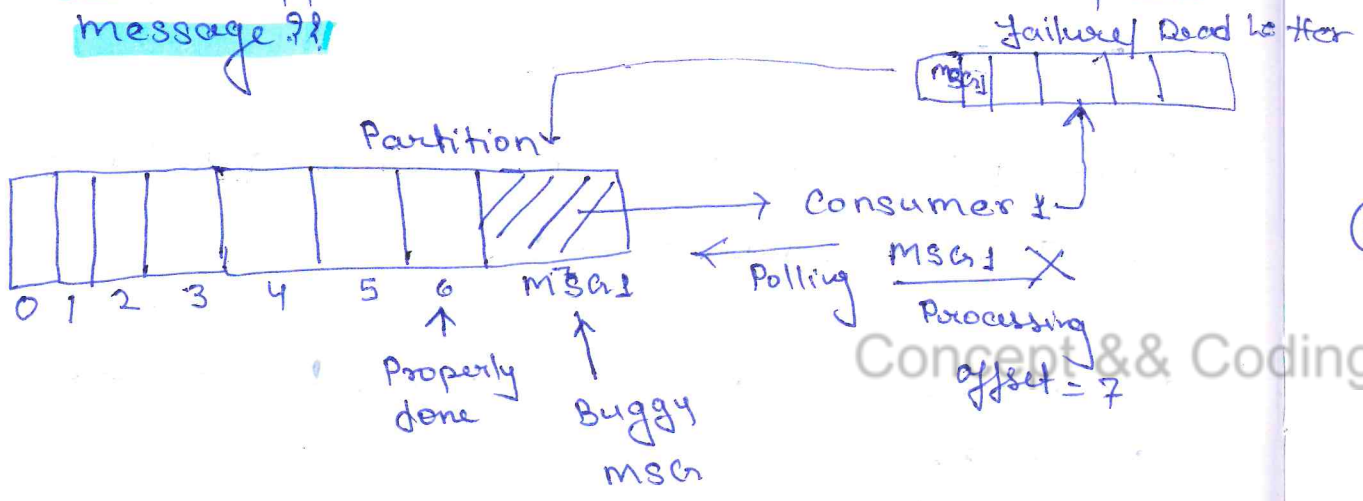
- Those where is not Replica is called as Leader.
- Those where is Replica is called as follower.

* So all Read/write → happens in through Leader
* Whenever Leader goes down then the Replica takes over and become a leader.



* follower always follow leader. Any new message comes in it takes it and also add in it's queue also. so that's how it keep on syncing itself from the leader.

(7) What happen when Consumer not able to process message ??



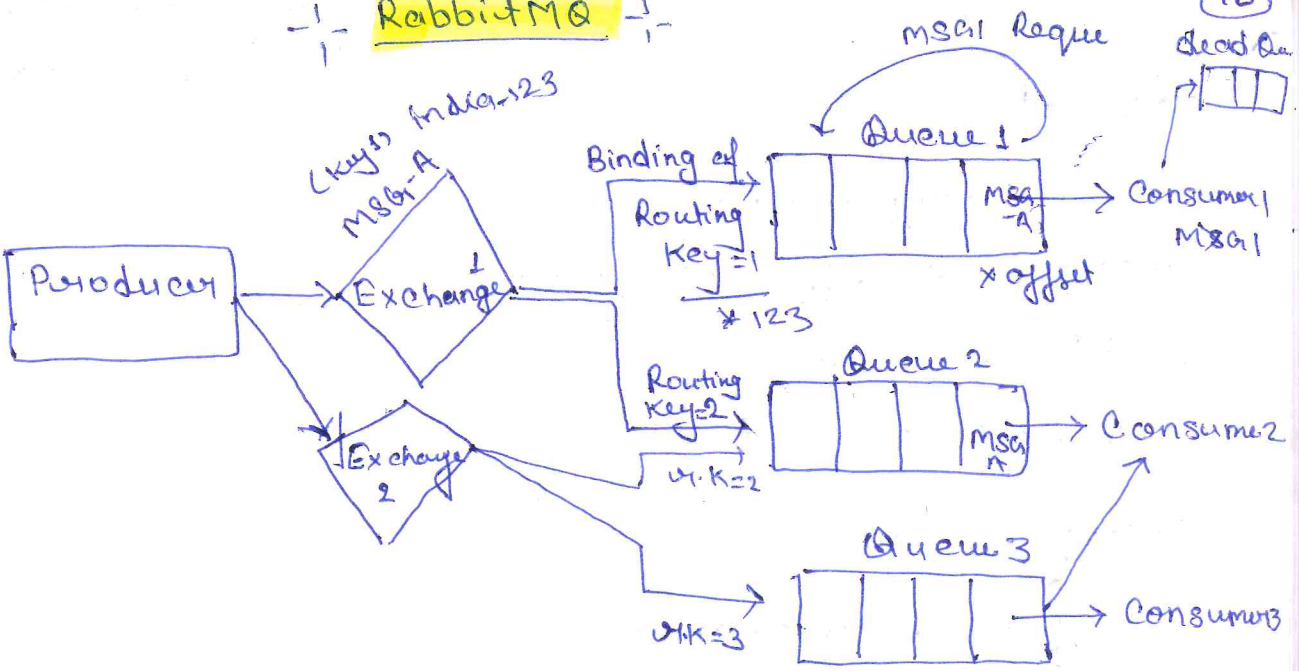
Concept && Coding

Commited offset = ~~6~~ 7

retry = 3 Every msg 3 thime Retry.

RabbitMQ	Kafka
Push base Approach	Pull based approach

RabbitMQ



other

Different type of Exchange:

1. **Fan Out Exchange** :- It will push the message to ~~the~~ all the queues which are associated with this exchang.

2. **Direct Exchange** :- (Exactly Maching)

   MSG key & Routing key.

3. **Topic Exchang** :- We can use [wild card]

   * 123

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* What happen Consumer One has reading a message and it's not able to process it. What it will do is it will reque to the back of queue it. put MSG 1 and Reque again.

Concept && Coding